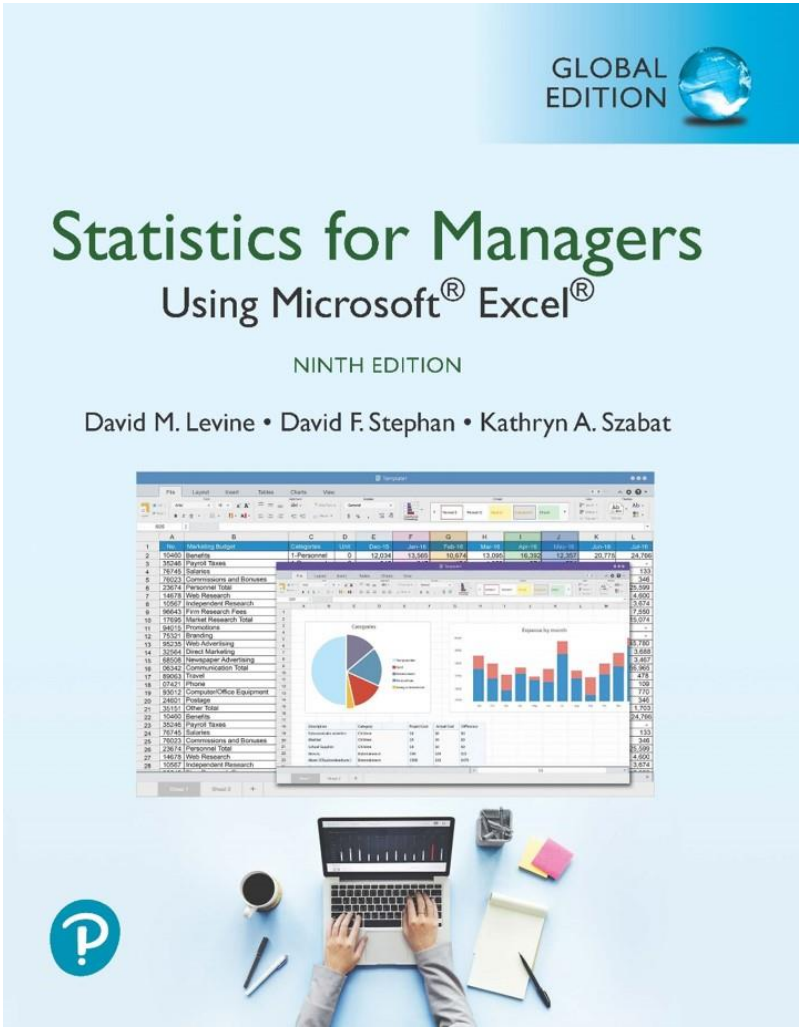


Chapter 9

Fundamentals of Hypothesis Testing: One-Sample Tests

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Objectives

In this chapter, you learn:

- The principles of hypothesis testing.
- How to use hypothesis testing to test a mean or proportion.
- To evaluate the assumptions of each hypothesis testing procedure and understand the consequences if assumptions are seriously violated.
- The pitfalls and ethical issues involved in hypothesis testing.
- How to avoid the pitfalls involved in hypothesis testing.

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What is an Hypothesis?

- A hypothesis is a claim (assertion) about a population parameter:

  - **Population mean (μ)**:

    Example: The average monthly allowance of tertiary students in the Western Cape is $\mathrm{R}4500$.

    $$\mu = 4500$$

  - **Population proportion (π)**:

    Example: The proportion of tertiary students in the Western Cape that own a car is 25%.

    $$\pi = 0.25$$

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Hypothesis Testing

- Previously, we sought to determine whether a sample mean is consistent with a known population mean.
- Here, we seek to validate a claim about the population using the sample mean.
- For this new goal, we use an inferential method called: *hypothesis testing*.

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Hypothesis Testing

- Steps for hypothesis testing:

1) State a straightforward claim about the population mean (μ). This claim is referred to as the *null hypothesis* (H_0).

2) State a mutually exclusive claim to the null hypothesis. Such a claim is referred to as the *alternative hypothesis* (H_1).

3) Begin with the assumption that H_0 is true.
   - **Recall**: innocent until proven guilty!

4) Examine the sample statistics to see if it better supports H_0 or H_1.

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The Null Hypothesis (H_0)

- States the claim or assertion to be validated.
- Assumes the status-quo scenario.
- Represents the current belief in a situation.
- May or may not be rejected.
  - Note: the null hypothesis is **NEVER** accepted.
- Always about a population parameter **NOT** about a sample statistic.

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The Null Hypothesis ($\mathbf{H_0}$)

Examples

- The average diameter of a bolt manufactured by a company for use in car engines is $30mm$.

$$\mathbf{H_0}: \boldsymbol{\mu = 30}$$

- A vending machine is programmed to dispense $240ml$ of hot chocolate per cup.

$$\mathbf{H_0}: \boldsymbol{\mu = 240}$$

- An airline advertised that its trip from Cape Town to Johannesburg is $110mins$ on average.

$$\mathbf{H_0}: \boldsymbol{\mu = 110}$$

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The Alternative Hypothesis (H_1)

- Is the opposite of the null hypothesis.
- Challenges the status-quo scenario.
- May or may not be proven.
- It is generally the hypothesis that the researcher intends to prove.

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The Alternative Hypothesis (H_1)

Examples

- The average diameter of bolts manufactured by a company for use in car engines is not $30mm$.

$$\mathbf{H_1: \mu \neq 30}$$

- The hot chocolate vending machine is programmed to dispense less than $240ml$ on average.

$$\mathbf{H_1: \mu < 240}$$

- The average travel time of an airline from Cape Town to Johannesburg is longer than $110mins$.

$$\mathbf{H_1: \mu > 110}$$

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The Hypothesis Testing Process

- Claim: *The population mean age is* 50.

$$H_0: \mu = 50 \quad H_1: \mu \neq 50$$

- Sample the population and calculate the sample mean.
- Suppose the sample mean value $\bar{x} = 20$.
- This is significantly less than the claimed mean population age of 50.
- If the null hypothesis were true, the probability of getting such a different sample mean would be very small, so reject the null hypothesis.
- In other words, getting a sample mean age of 20 is so unlikely if the population mean is 50.
- Conclude that the population mean could not have been 50.

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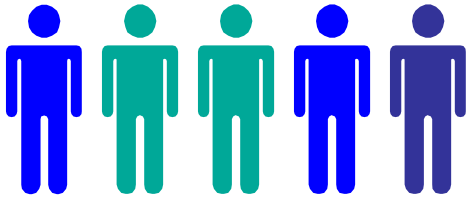
Concepts of Hypothesis Testing

Claim: the population mean age is 50 years. That is, null hypothesis, H_0: $\mu = 50$.



Population

Select a random sample



Sample

Suppose the sample mean age is 20. That is, $\bar{x} = 20$.

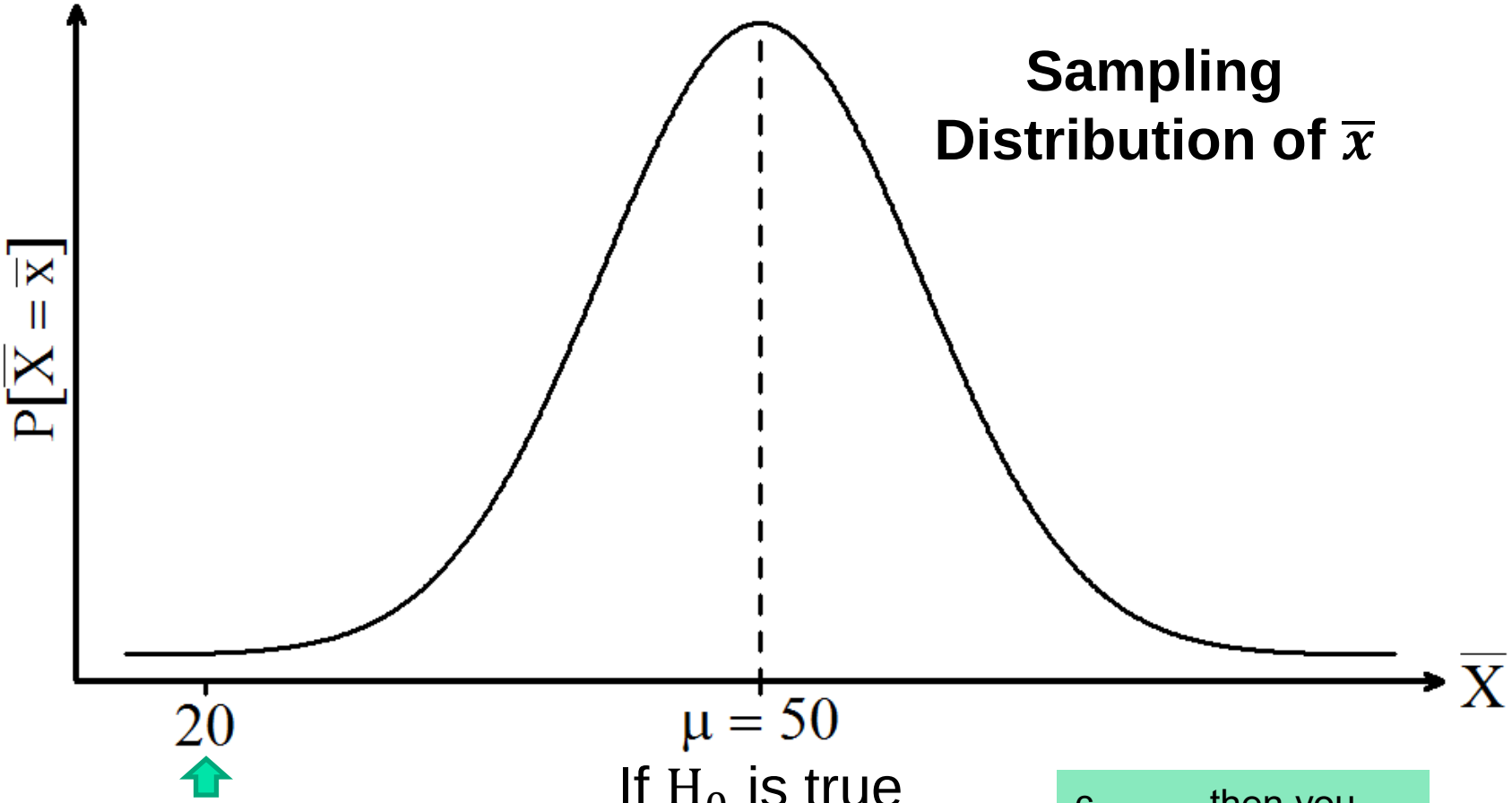
Is $\bar{x} = 20$ likely if $\mu = 50$?

If not likely, REJECT null hypothesis

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The Hypothesis Testing Process



a. If it is unlikely that you would get a sample mean of this value …

b. … when in fact this was the population mean …

c. … then you reject the null hypothesis that $\mu = 50$.

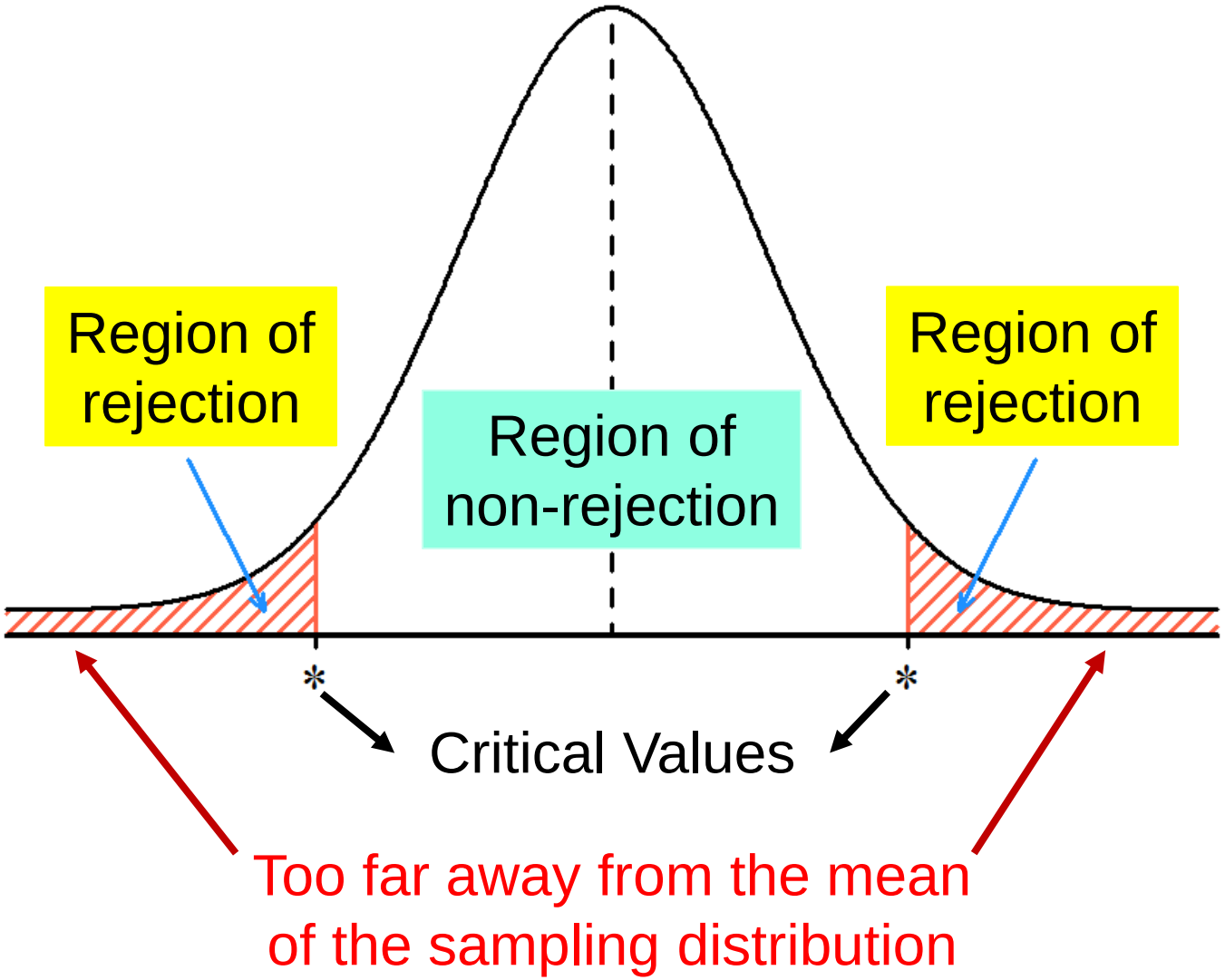
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The Test Statistic and Critical Values

- If the sample mean is **close** to the stated population mean, H_0 is **not rejected**.
- If the sample mean is **far** from the stated population mean, H_0 is **rejected**.
- How far is "far enough" or how close is "close enough" for a decision about H_0 to be made?
  - The *critical value* of a *test statistic* helps to answer this decision-making question.
  - For hypothesis test with respect to the population mean, the test statistic is the sample mean.

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The Test Statistic and Critical Values



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Risks in Decision Making Using Hypothesis Testing

- Type I Error:
  - Reject a true null hypothesis (H_0).
  - A Type I error is a "false alarm".
  - The probability of a Type I error is often denoted by $\boldsymbol{\alpha}$.
    - Called "level of significance" of the test.
    - Specified by the researcher in advance.

- Type II Error:
  - Failure to reject a false null hypothesis (H_0).
  - Type II error represents a "missed opportunity".
  - The probability of a Type II error is often denoted by $\boldsymbol{\beta}$.

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Risks in Decision Making Using Hypothesis Testing

Possible Hypothesis Test Outcomes

	Actual Situation	
Decision	H_0 is True	H_0 is False
Do not reject H_0	Correct decision $\text{confidence} = 1 - \alpha$	Type II error $P(\text{Type II Error}) = \beta$
Reject H_0	Type I error $P(\text{Type I Error}) = \alpha$	Correct decision $\text{power} = 1 - \beta$

- The *confidence coefficient* $(1 - \alpha)$ is the probability of not rejecting a true H_0.
- The *confidence level* of a hypothesis test is calculated as $(1 - \alpha) \times 100\%$.
- The *power of a statistical test* $(1 - \beta)$ is the probability of rejecting a false H_0.

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Type I and Type II Errors Relationship

- Type I and Type II errors cannot happen at the same time.
- A Type I error can only occur if H_0 is true.
- A Type II error can only occur if H_0 is false.

If the probability of Type I error (α) increases 

then the probability of Type II error (β) decreases 

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Factors That Influence Type II Error

- All else equal:
  - β increases when the difference between hypothesised parameter and its true value decreases.
  - β increases when $\mathrm{P}(\text{Type I Error}) = \alpha$ decreases.
  - β increases when variance (σ^2) increases.
  - β increases when sample size (n) decreases.

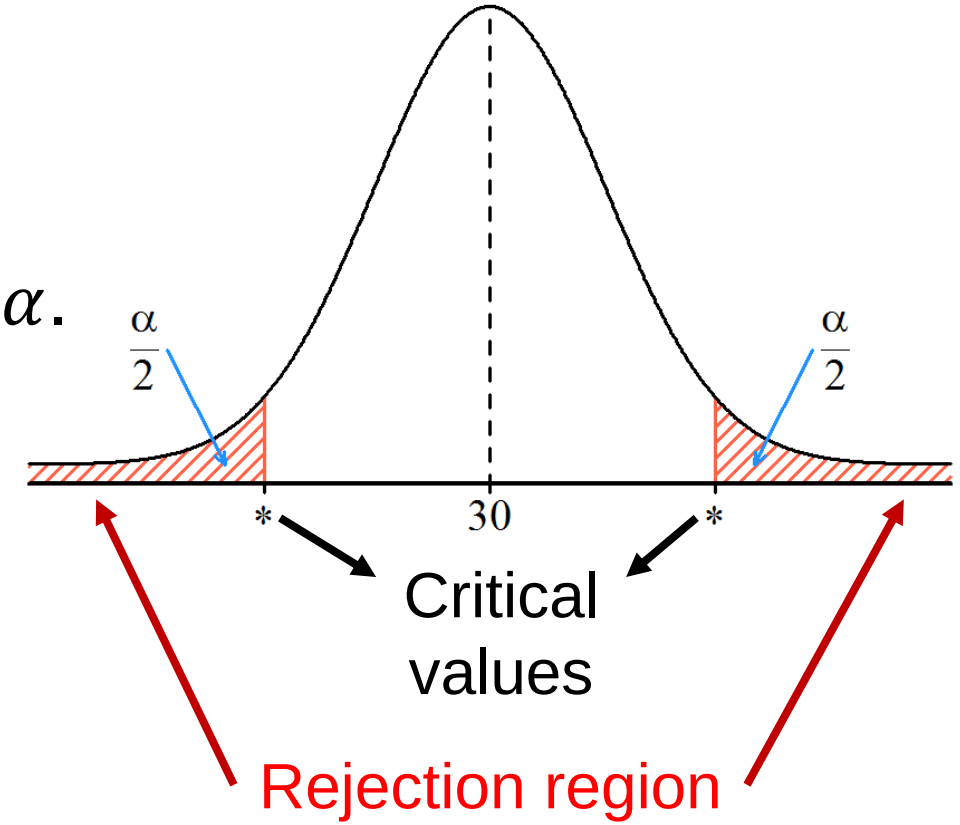
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Level of Significance and Rejection Region

Two-tail Test

- H_0: $\mu = 30$.
- H_1: $\mu \neq 30$.
- Significance level $= \alpha$.

This is a two-tail test because there is a rejection region at both tails.



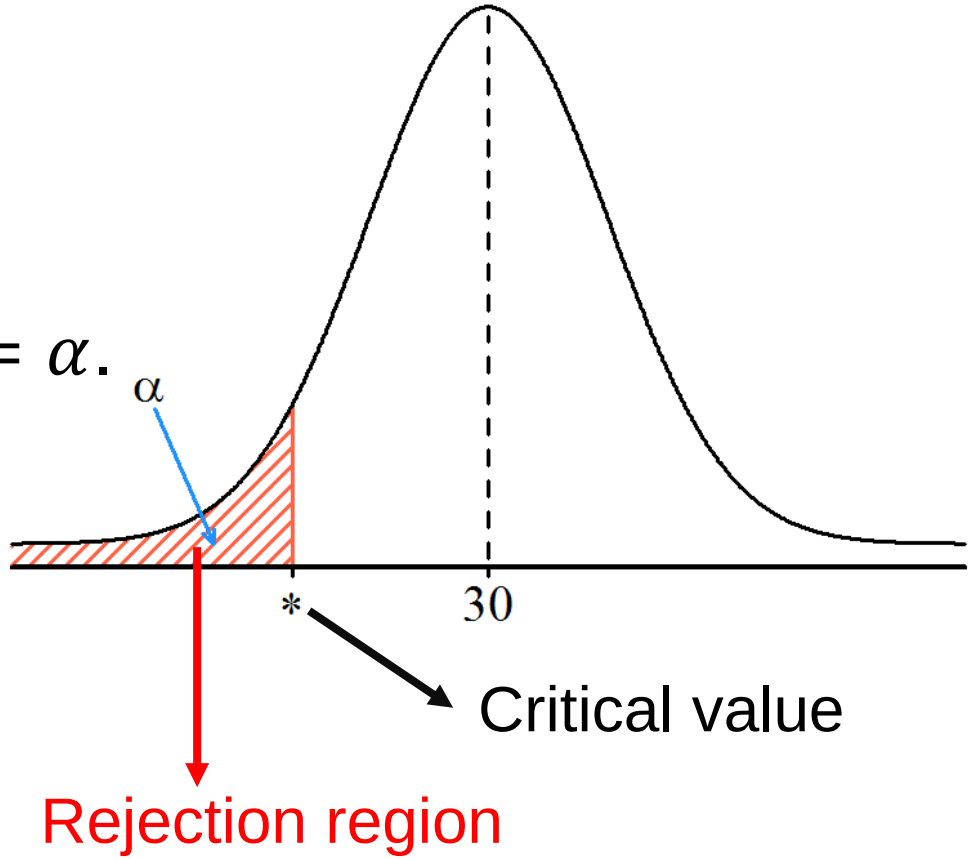
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Level of Significance and Rejection Region

One-tail Test

- H_0: $\mu \geq 30$.
- H_1: $\mu < 30$.
- Significance level $= \alpha$.

This is a one-tail test because there is a rejection region at the left tail.



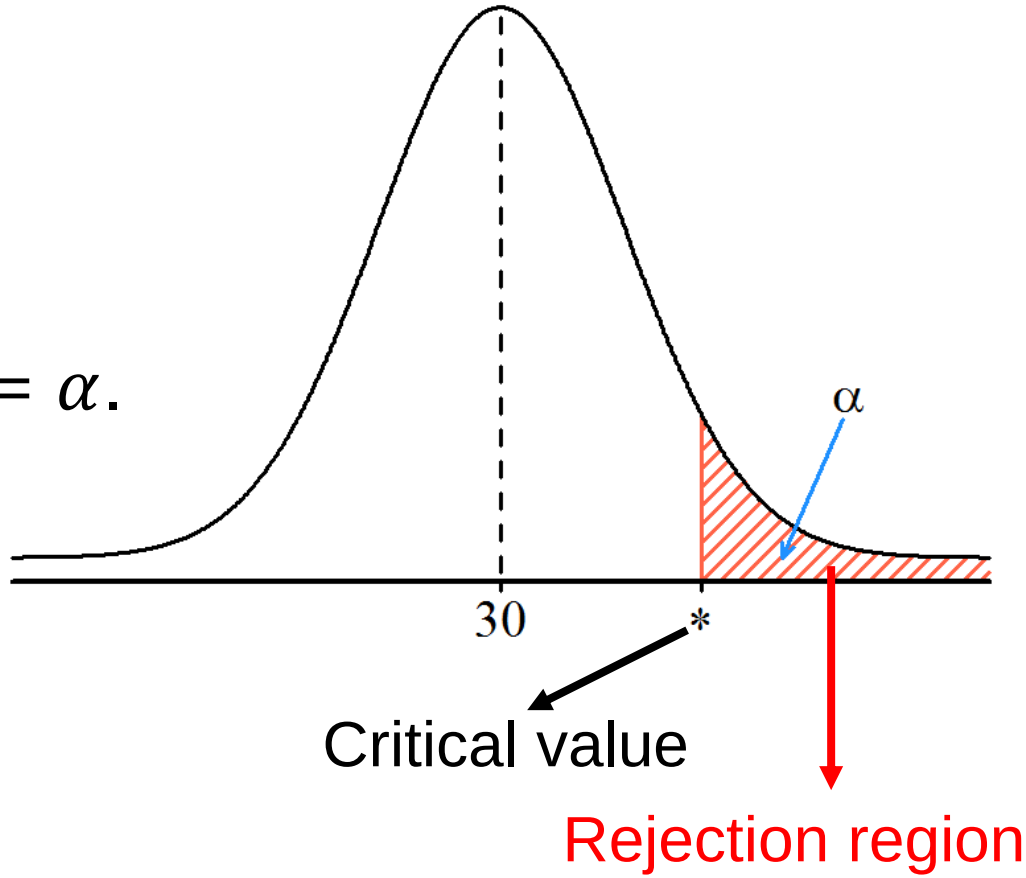
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Level of Significance and Rejection Region

One-tail Test

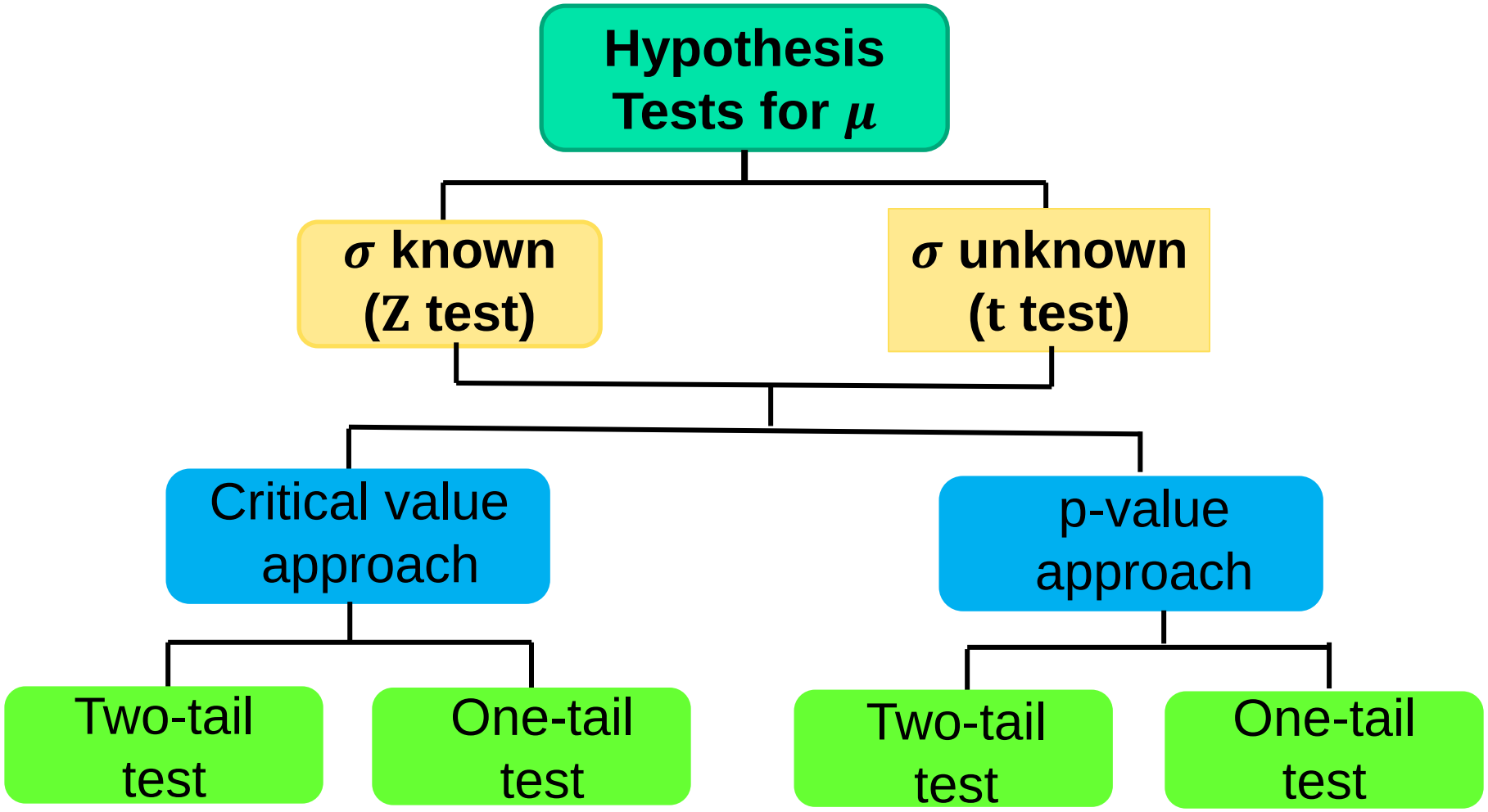
- H_0: $\mu \leq 30$.
- H_1: $\mu > 30$.
- Significance level $= \alpha$.

This is a one-tail test because there is a rejection region at the right tail.



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Hypothesis Tests for the Mean



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Z Test of Hypothesis for the Mean (σ Known)

Test Statistic

- The test statistic when the population standard deviation (σ) is known can be expressed as follows.

$$z_{stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

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Z Test of Hypothesis for the Mean (σ Known)

Critical Value Approach

- Convert sample statistic $(\bar{x})$ to test statistic (Z_{stat}).
- Determine the critical Z -value for a specified level of significance (α) from a table or by using Microsoft Excel.
- Decision Rule: If the test statistic falls in the rejection region, reject H_0. Otherwise, do not reject H_0.

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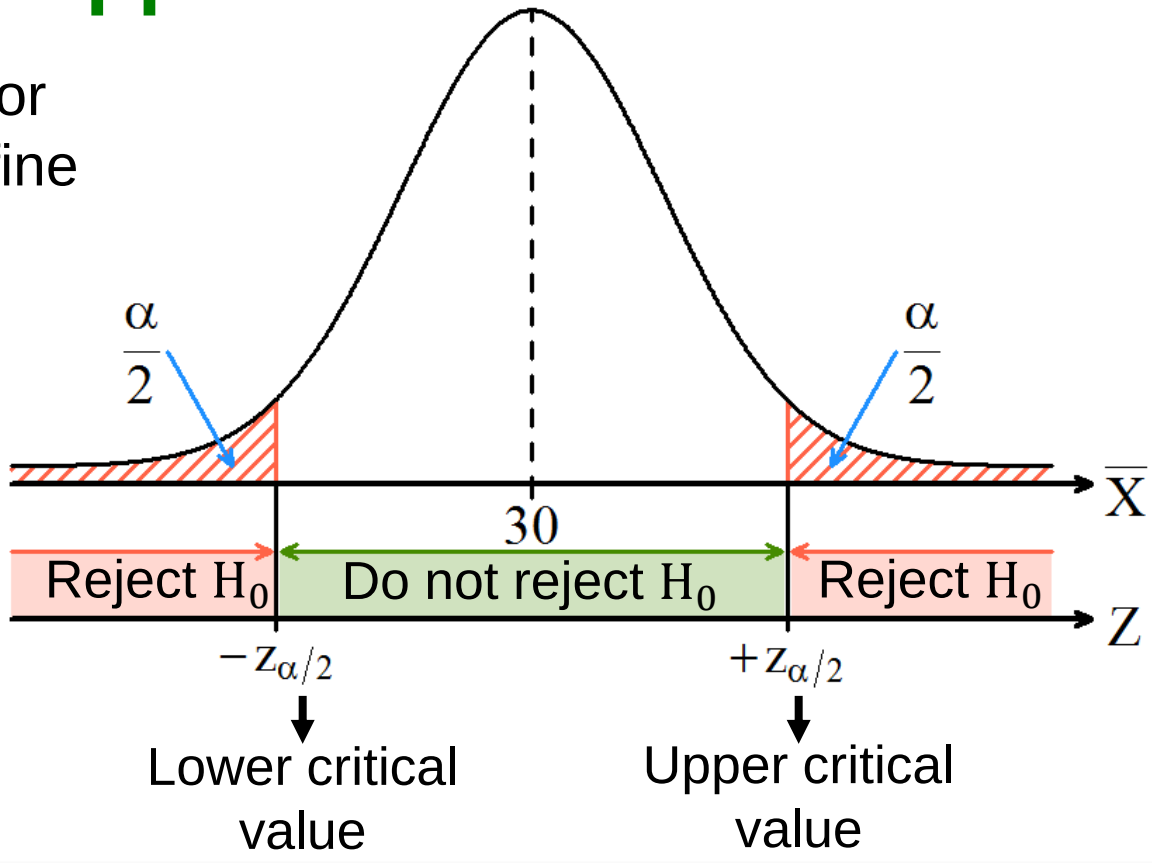
Z Test of Hypothesis for the Mean (σ Known)

Critical Value Approach: Two-tail Test

- There are two cut-off (or critical) values that define the rejection regions.

$\mathbf{H_0}: \boldsymbol{\mu} = \mathbf{30}$

$\mathbf{H_1}: \boldsymbol{\mu} \neq \mathbf{30}$



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Steps in the Critical Value Approach to Hypothesis Testing

1. State the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. Choose the level of significance (α) and the sample size (n). The level of significance is based on the relative importance of Type I and Type II errors in the problem.
3. Determine the appropriate test statistic and sampling distribution.
4. Determine the critical values that divide the rejection and non-rejection regions.

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Steps in the Critical Value Approach to Hypothesis Testing

5. Collect the sample data, organise the results and compute the value of the test statistic.

6. Make the statistical decision, determine whether the assumptions are valid and state the managerial conclusion in the context of the theory, claim or assertion being tested.

   - If the test statistic falls in the non-rejection region, you do not reject the null hypothesis H_0.
   - If the test statistic falls in the rejection region, reject the null hypothesis.

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Critical Value Approach to Hypothesis Testing
Example (Two-Tail Test)

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume $\sigma = 0.80$.

1. State the appropriate null and alternative hypotheses.
   - H_0: $\mu = 30$ vs H_1: $\mu \neq 30$ (this is a two-tail test).

2. Specify the desired level of significance and sample size.
   - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.

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Critical Value Approach to Hypothesis Testing

Example (Two-Tail Test)

3. Determine the appropriate technique.
   - σ is assumed known. So, this is a Z test.

4. Determine the critical values.
   - With $\alpha = 0.05$ for a two-tail test the critical Z values are ± 1.96.

Cumulative Standardised Normal Probabilities

Z	0.05	0.06	0.07
-2.1	0.0158	0.0154	0.0150
-2.0	0.0202	0.0197	0.0192
-1.9	0.0256	**0.0250**	0.0244
-1.8	0.0322	0.0314	0.0307
-1.7	0.0401	0.0392	0.0384

Z	0.05	0.06	0.07
1.7	0.9599	0.9608	0.9616
1.8	0.9678	0.9686	0.9693
1.9	0.9744	**0.9750**	0.9756
2.0	0.9798	0.9803	0.9808
2.1	0.9842	0.9846	0.9850

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Critical Value Approach to Hypothesis Testing
Example (Two-Tail Test)

5. Collect the data and compute the test statistic.
   - Suppose that the sample results are the following.

$n = 100$ $\quad \bar{x} = 29.84$ $\quad \sigma = 0.80$ (assume known)

Therefore, the test statistic is:

$$z_{stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{29.84 - 30}{0.80/\sqrt{100}} = \frac{-0.16}{0.08} = -2.0$$

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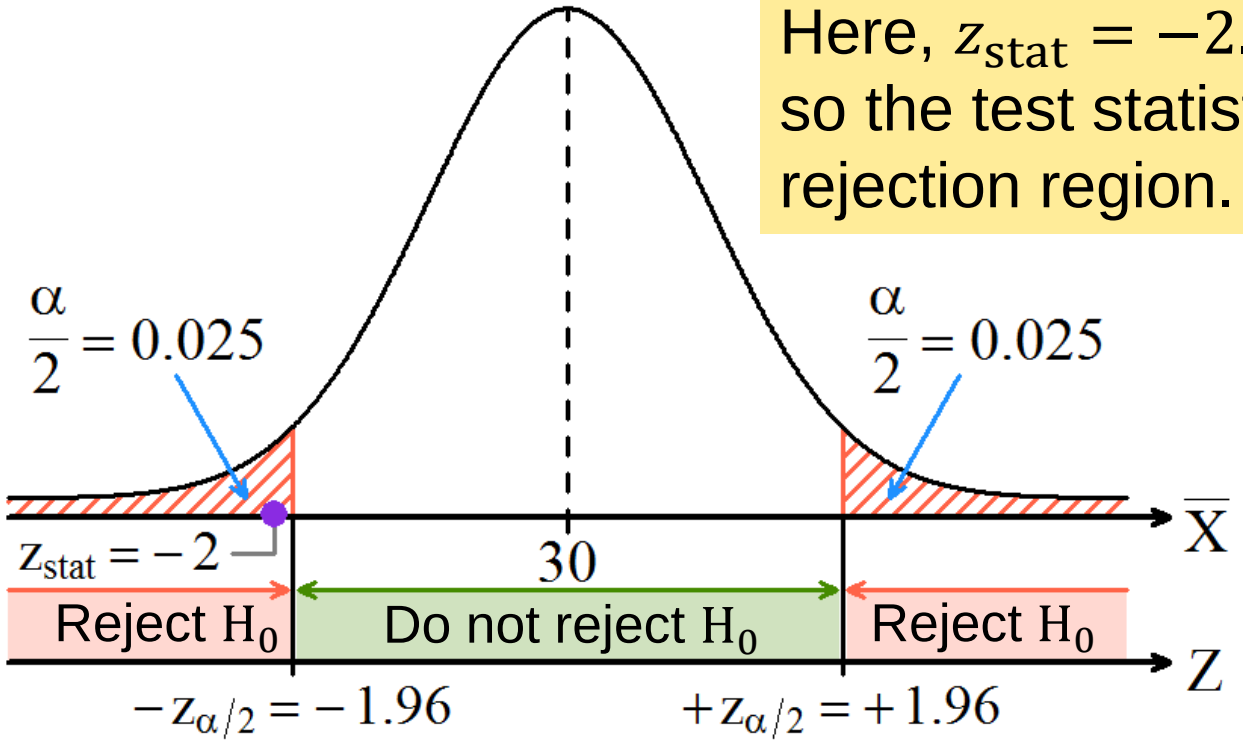
Critical Value Approach to Hypothesis Testing

Example (Two-Tail Test)

6. Is the test statistic in the rejection region?

- Reject H_0 if $z_{stat} < -1.96$ or if $z_{stat} > +1.96$. Otherwise, do no reject H_0.

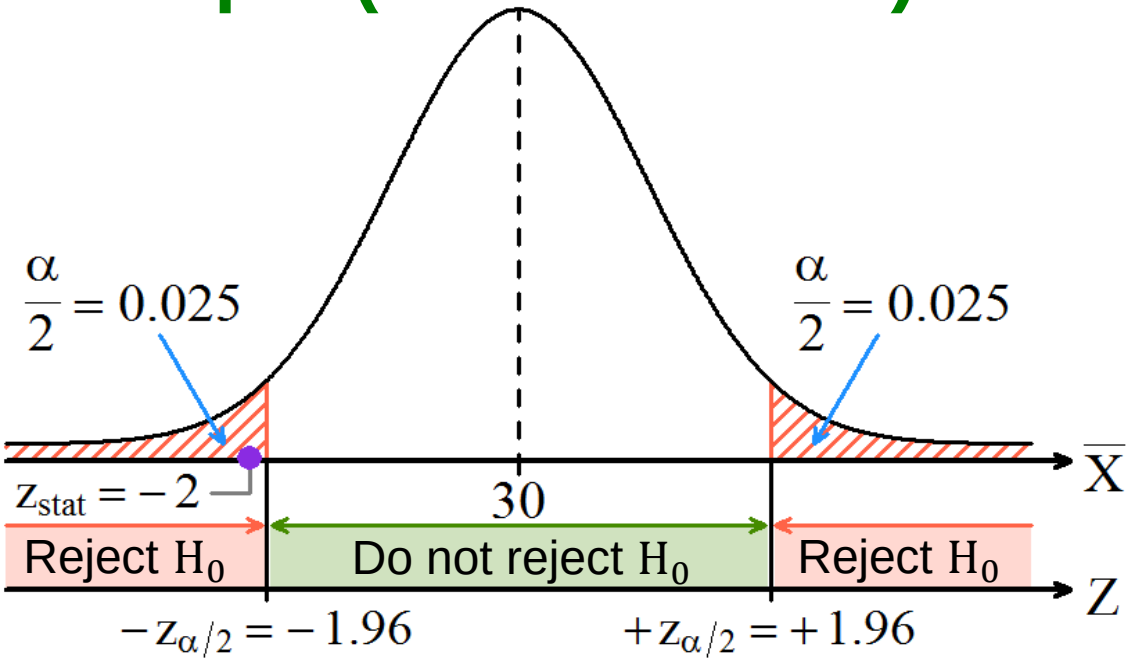
Here, $z_{stat} = -2.0 < -1.96$, so the test statistic is in the rejection region.



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Critical Value Approach to Hypothesis Testing

Example (Two-Tail Test)



7. Reach a decision and interpret the result.

   - Since $z_{stat} = -2.0 < -1.96$, reject the null hypothesis and conclude that there is sufficient evidence that the mean diameter of a manufactured bolt is not equal to $30mm$.

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p-value Approach to Hypothesis Testing

- p-value: probability of obtaining a test statistic equal to or more extreme than the observed sample value, given that H_0 is true.
  - The p-value is also called the observed level of significance.
  - It is the smallest value of α for which H_0 can be rejected.
- Interpreting the p-value: compare the p-value with α.
  - If the p-value $< \alpha$, reject H_0.
  - If the p-value $\geq \alpha$, do not reject H_0.

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The 5 Step p-value Approach to Hypothesis Testing

1. State the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. Choose the level of significance (α) and the sample size (n).
   - The level of significance is based on the relative importance of the risks of Type I and Type II errors.
3. Determine the appropriate test statistic and sampling distribution.
4. Collect the sample data, compute the value of the test statistic and the p-value.
5. Make the statistical decision and state the managerial conclusion in the context of the theory, claim or assertion being tested.
   - If the p-value is less than α, reject H_0.

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p-value Approach to Hypothesis Testing
Example (Two-Tail Test)

Test the claim that the true mean diameter of a manufactured bolt is 30mm. Assume $\sigma = 0.80$.

1. State the appropriate null and alternative hypotheses.
   - H_0: $\mu = 30$ vs H_1: $\mu \neq 30$ (this is a two-tail test).

2. Specify the desired level of significance and sample size.
   - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test.

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p-value Approach to Hypothesis Testing
Example (Two-Tail Test)

3. Determine the appropriate technique.
   - σ is assumed known. So, this is a Z test.

4a. Collect the data and compute the test statistic.
   - Suppose that the sample results are the following.

$n = 100$ $\quad \bar{x} = 29.84$ $\quad \sigma = 0.80$ (assume known)

Therefore, the test statistic is:

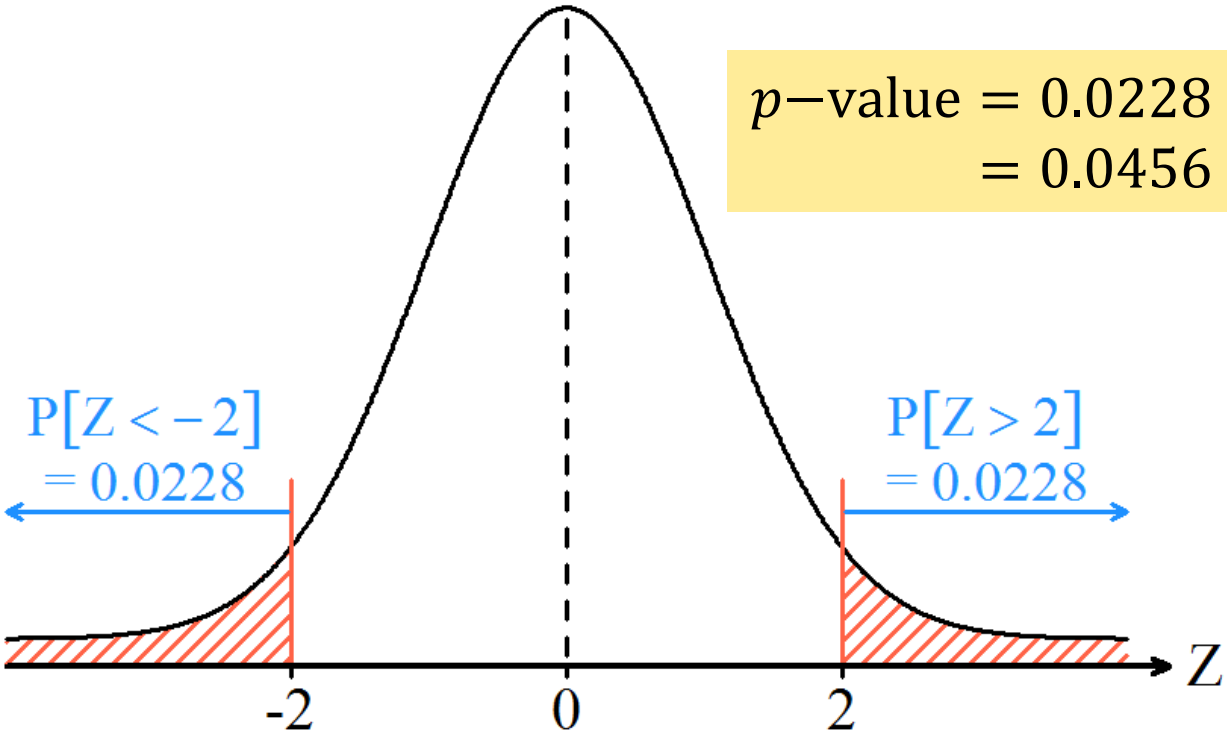
$$z_{stat} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{29.84 - 30}{0.80/\sqrt{100}} = \frac{-0.16}{0.08} = -2.0$$

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p-value Approach to Hypothesis Testing
Example (Two-Tail Test)

4b. Calculate the p-value.

- How likely is it to get a z_{stat} of -2 (or a value further from the mean $[= 0]$, in either direction) if H_0 is true?



$$p\text{−value} = 0.0228 + 0.0228 = 0.0456$$

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p-value Approach to Hypothesis Testing
Example (Two-Tail Test)

5a. Is the p-value $< \alpha$.

- Since p-value $= 0.0456 < \alpha = 0.05$, reject H_0.

5b. State the managerial conclusion in the context of the situation.

- There is sufficient evidence to conclude that the mean diameter of a manufactured bolt is not equal to $30mm$.

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Two-Tail Hypothesis Test for Mean

p-value Approach (Excel)

	A	B	C
1	**Data**		**Excel Commands**
2	Null Hypothesis (H_0) $\mu =$	30.00	
3	Alternative Hypothesis (H_1) $\mu \neq$	30.00	
4	Significance Level (α)	0.05	
5	Sample Size (n)	100.00	
6	Sample Mean $(\bar{x})$	29.84	
7	Population Standard Deviation (σ)	0.80	
8			
9	**Intermediate Calculations**		
10	Standard Error of the Mean	0.0800	= B7 / SQRT(B5)
11	z test Statistic	-2.0000	= (B6 - B2) / B10
12			
13	**Two-Tail Hypothesis Test**		
14	Lower Critical Value	-1.9600	= NORM.S.INV(B4 / 2)
15	Upper Critical Value	1.9600	= NORM.S.INV(1 - B4 / 2)
16	p-value	0.0455	= (1 - NORM.S.DIST(ABS(B11), TRUE)) * 2
17	Decision	**Reject null hypothesis**	= IF(B16 < B4, "Reject null hypothesis", "Do not reject null hypothesis")

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Connection Between Two-Tail Z Tests and Confidence Intervals

Recall,

$$\overline{\mathrm{X}} \sim \mathrm{N}(\mu, \sigma^2/n)$$

Implies that,

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

So that,

$$\mathrm{P}[-1.96 < \mathrm{Z} < 1.96] = 0.95$$

$$\mathrm{P}\left[-1.96 < \frac{\overline{\mathrm{X}} - \mu}{\sigma/\sqrt{n}} < 1.96\right] = 0.95$$

$$\mathrm{P}\left[\bar{x} - 1.96 \times \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96 \times \frac{\sigma}{\sqrt{n}}\right] = 0.95$$

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Connection Between Two-Tail Z Tests and Confidence Intervals

- For $\bar{x} = 29.84$, $\sigma = 0.80$, $n = 100$ and $\alpha = 5\%$, the 95% confidence interval for μ is:

$$\bar{x} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$29.84 - 1.96 \times \frac{0.8}{\sqrt{100}} \leq \mu \leq 29.84 + 1.96 \times \frac{0.8}{\sqrt{100}}$$

$$29.6832 \leq \mu \leq 29.9968$$

- Since this interval does not contain the hypothesised mean (30), we reject the null hypothesis at $\alpha = 0.05$.

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Do You Ever Know σ?

- Probably not!
- In virtually all real-world business situations, σ is not known.
- If there is a situation where σ is known, then μ should also be known because μ is needed to compute σ.
- If we truly know μ, there would be no need to gather a sample to estimate it.

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Hypothesis Testing for the Mean (σ Unknown)

- If the population standard deviation (σ) is unknown, you use the sample standard deviation (s) instead.
- Because of this change, it is important to replace the Z distribution with the ***t* distribution** when testing null hypothesis about the mean.
- When using the ***t* distribution** you must assume that the population you are sampling from follows a normal distribution.
- All other steps, concepts and conclusions from the Z distribution remain the same for the ***t* distribution**.

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t Test of Hypothesis for the Mean (σ Unknown)

- In order to use the t test, it is important to assume that the population distribution of the variable of interest is normal.
- As long as the sample size is not very small and the population is not very skewed, the t test can be used.
- Recall that to evaluate the normality assumption:
  - Determine how closely sample statistics match the normal distribution's theoretical properties.
  - Assess symmetry of the variable graphically with histogram or stem-and-leaf plot or boxplot.
  - Assess linearity of the normal probability plot.

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t Test of Hypothesis for the Mean (σ Unknown)

Test Statistic

- When the population standard deviation is unknown, the sample standard deviation is used to approximate it. The test statistic can then be expressed as follows.

$$t_{stat} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

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t Test of Hypothesis for the Mean (σ Unknown)

Critical Value Approach: Two-tail Test

Example

The mean cost of a large-sized pizza in the Western Cape is said to be $\mathrm{R}168$. To determine if this is true, a random sample of 25 pizza outlets (from separate brands) is taken and this resulted in a sample mean $(\bar{x})$ of $\mathrm{R}172.50$ and a sample standard deviation (s) of $\mathrm{R}15.40$. Test the appropriate hypotheses at $\alpha = 0.05$. (Assume that the population distribution is normal)

$\mathbf{H_0}: \boldsymbol{\mu = 168}$ $\mathbf{H_1}: \boldsymbol{\mu \neq 168}$

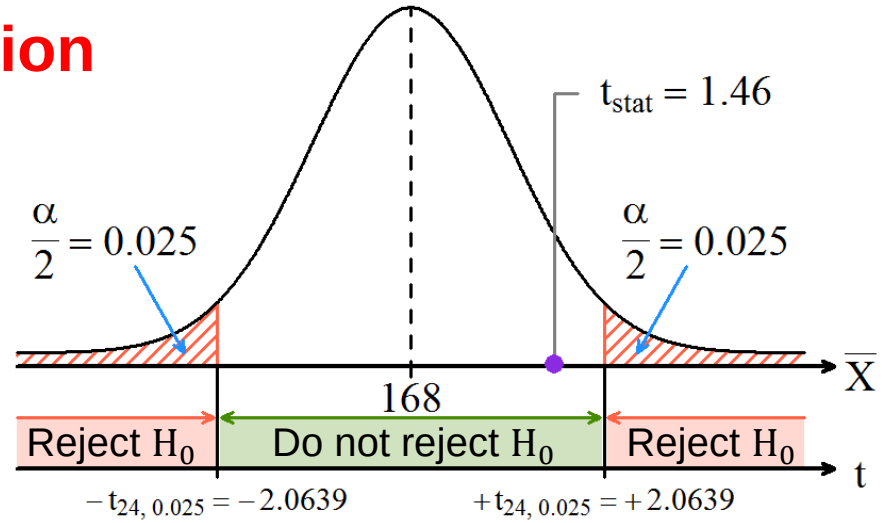
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t Test of Hypothesis for the Mean (σ Unknown)

Critical Value Approach: Two-tail Test

Solution

- $\alpha = 0.05.$
- $n = 25.$
- Degrees of Freedom $df = 25 - 1 = 24.$
- σ is unknown, so use a ***t* statistic**.
- Critical value: $\pm t_{24,0.025} = \pm 2.0639.$



$$t_{stat} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{172.50 - 168}{15.40/\sqrt{25}} = 1.46$$

Do not reject H_0: insufficient evidence that the true mean cost of a large pizza is different from R168.

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Two-Tail Hypothesis (t) Test for Mean

σ Unknown | p-value Approach (Excel)

	A	B	C
1	**Data**		**Excel Commands**
2	Null Hypothesis (H_0) $\mu =$	168.00	
3	Alternative Hypothesis (H_1) $\mu \neq$	168.00	
4	Significance Level (α)	0.05	
5	Sample Size (n)	25.00	
6	Sample Mean ($\bar{x}$)	172.50	
7	Sample Standard Deviation (s)	15.40	
8			
9	**Intermediate Calculations**		
10	Standard Error of the Mean	3.0800	= B7 / SQRT(B5)
11	Degrees of Freedom	24.0000	= B5 - 1
12	t test Statistic	1.4610	= (B6 - B2) / B10
13			
14	**Two-Tail Hypothesis Test**		
15	Lower Critical Value	-2.0639	= T.INV(B4 / 2, B11)
16	Upper Critical Value	2.0639	= T.INV(1 - B4 / 2, B11)
17	p-value	0.1570	= T.DIST.2T(ABS(B12), B11)
18	Decision	**Do not reject null hypothesis**	= IF(B17<B4, "Reject null hypothesis", "Do not reject null hypothesis")

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Connection Between Two-Tail t Tests and Confidence Intervals

Recall that when population variance is unknown,

$$\overline{\mathrm{X}} \dot{\sim} \mathrm{N}(\mu, s^2/n)$$

Implies that,

$$t = \frac{\overline{\mathrm{X}} - \mu}{s/\sqrt{n}}$$

So that,

$$\mathrm{P}\left[t_{n-1,0.025} < \mathrm{t} < t_{n-1,0.975}\right] = 0.95$$

$$\mathrm{P}\left[t_{n-1,0.025} < \frac{\overline{\mathrm{X}} - \mu}{s/\sqrt{n}} < t_{n-1,0.975}\right] = 0.95$$

$$\mathrm{P}\left[\bar{x} - t_{n-1,0.025} \times \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1,0.975} \times \frac{s}{\sqrt{n}}\right] = 0.95$$

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Connection Between Two-Tail t Tests and Confidence Intervals

- For $\bar{x} = 172.5$, $s = 15.40$, $n = 25$ and $\alpha = 5\%$, the 95% confidence interval for μ is:

$$\bar{x} - t_{n-1,\alpha/2} \times \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{n-1,\alpha/2} \times \frac{s}{\sqrt{n}}$$

$$172.5 - 2.0639 \times \frac{15.4}{\sqrt{25}} \leq \mu \leq 172.5 + 2.0639 \times \frac{15.4}{\sqrt{25}}$$

$$166.1432 \leq \mu \leq 178.8568$$

- Since this interval contains the hypothesised mean (168), we fail to reject the null hypothesis at $\alpha = 0.05$.

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One-Tail Hypothesis Tests

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One-Tail Hypothesis Tests for the Mean

- In many cases, the alternative hypothesis focuses on a particular direction.

Case 1: Lower-Tail Test

$\mathbf{H_0}: \boldsymbol{\mu} \geq \mathbf{3}$
$\mathbf{H_1}: \boldsymbol{\mu} < \mathbf{3}$

This is a lower-tail test because, relatively, the alternative hypothesis (H_1) focuses on the left side of the number line. That is, below the mean (μ) of 3.

Case 2: Upper-Tail Test

$\mathbf{H_0}: \boldsymbol{\mu} \leq \mathbf{3}$
$\mathbf{H_1}: \boldsymbol{\mu} > \mathbf{3}$

This is a upper-tail test because, relatively, the alternative hypothesis (H_1) focuses on the right side of the number line. That is, above the mean (μ) of 3.

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One-Tail Hypothesis Tests for the Mean

- It is common practice to approximate the exact inequality sign of the null hypothesis in one-tail tests with an equal sign.

Case 1: Lower-Tail Test

$$\begin{aligned} H_0&: \mu \geq 3 \\ H_1&: \mu < 3 \end{aligned} \quad \approx \quad \begin{aligned} H_0&: \mu = 3 \\ H_1&: \mu < 3 \end{aligned}$$

Case 2: Upper-Tail Test

$$\begin{aligned} H_0&: \mu \leq 3 \\ H_1&: \mu > 3 \end{aligned} \quad \approx \quad \begin{aligned} H_0&: \mu = 3 \\ H_1&: \mu > 3 \end{aligned}$$

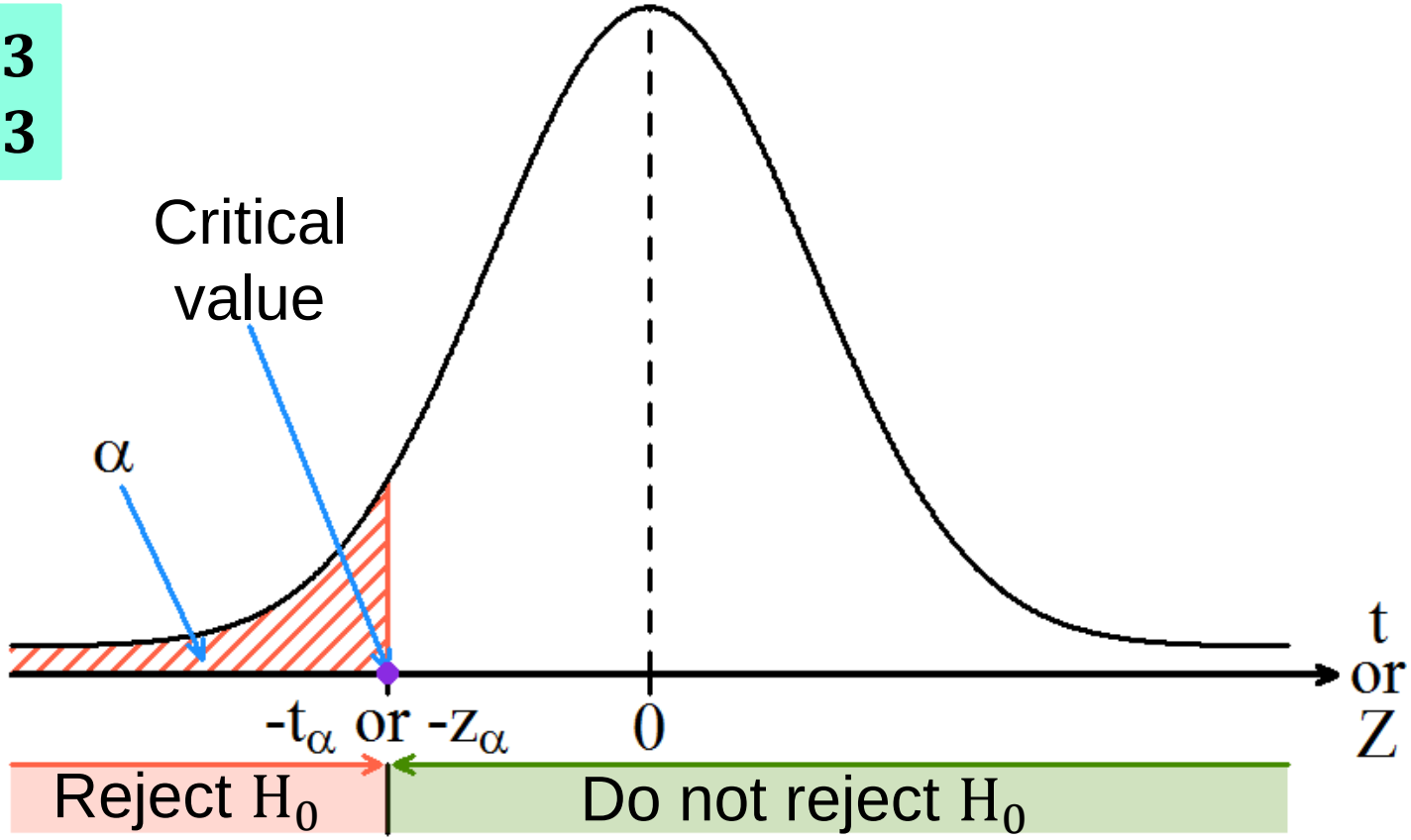
- For simplicity, the null hypothesis approximations specified above will be adopted in SDS188.

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Lower-Tail Test

- There is only one critical value since the rejection area is in only one tail – on the left side.

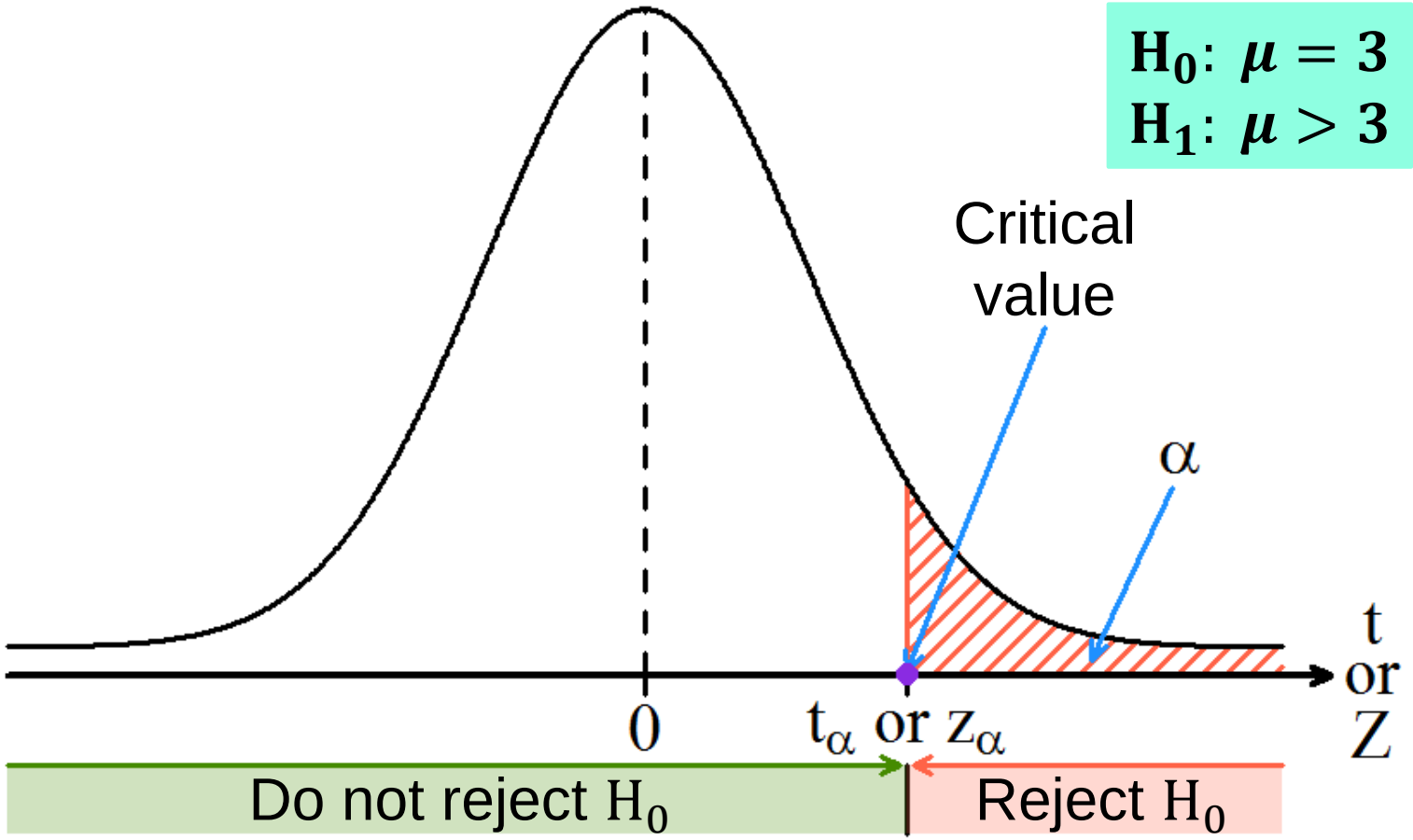
$\mathbf{H_0}: \boldsymbol{\mu} = \mathbf{3}$
$\mathbf{H_1}: \boldsymbol{\mu} < \mathbf{3}$



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Upper-Tail Test

- There is only one critical value since the rejection area is in only one tail – on the right side.



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One-Tail Hypothesis Test for Mean

Upper-Tail | σ Unknown | Critical Value Approach

Example

A phone industry manager thinks that customer monthly phone bills have increased and now average over $\mathrm{R}52$ per month. The company wishes to test the claim. Subsequently, data from 25 customers were collected and this resulted in a sample mean $(\bar{x})$ of $\mathrm{R}53.10$ and a sample standard deviation (s) of $\mathrm{R}10.00$. Test the appropriate hypotheses at a 10% significance level. (Assume that the population distribution is normal)

$\mathbf{H_0}$: $\boldsymbol{\mu = 52}$ the mean monthly bill remains $\mathrm{R}52$ per month.

$\mathbf{H_1}$: $\boldsymbol{\mu > 52}$ the mean is greater than $\mathrm{R}52$ per month.

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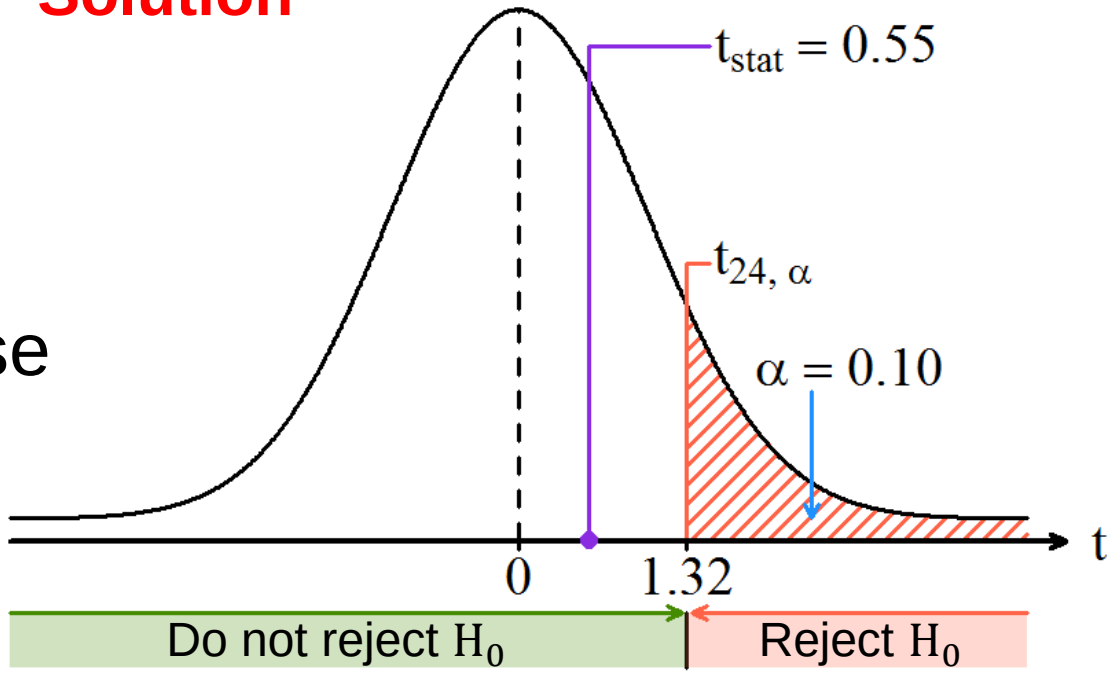
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One-Tail Hypothesis Test for Mean

Upper-Tail | σ Unknown | Critical Value Approach

Solution

- $\alpha = 0.10.$
- $n = 25.$
- σ is unknown, so use a ***t* statistic**.
- Degrees of Freedom $df = 25 - 1 = 24.$
- Critical value: $t_{24,0.10} = 1.3178.$



$$t_{stat} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{53.10 - 52}{10/\sqrt{25}} = 0.55$$

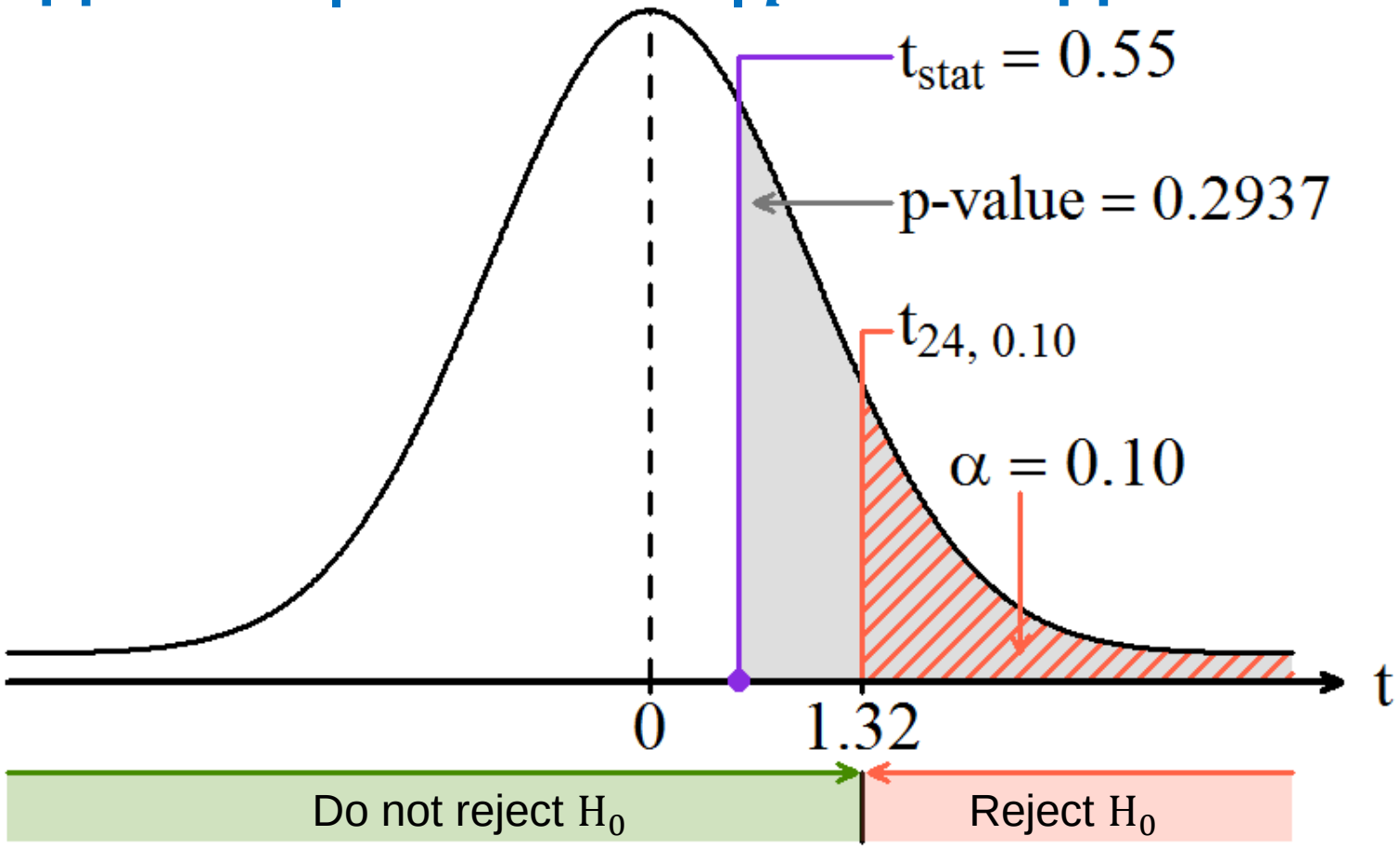
Do not reject H_0: insufficient evidence that the true mean monthly phone bill is greater than R52.

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One-Tail Hypothesis Test for Mean

Upper-Tail | σ Unknown | p-value Approach



Do not reject H_0: insufficient evidence that the true mean monthly phone bill is greater than R52 since p-value $\approx 0.2937 > \alpha = 0.10$.

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One-Tail Hypothesis (t) Test for Mean

Upper-Tail | σ Unknown | p-value Approach (Excel)

	A	B	C
1	**Data**		**Excel Commands**
2	Null Hypothesis (H_0) $\mu =$	52.00	
3	Alternative Hypothesis (H_1) $\mu >$	52.00	
4	Significance Level (α)	0.10	
5	Sample Size (n)	25.00	
6	Sample Mean $(\bar{x})$	53.10	
7	Sample Standard Deviation (s)	10.00	
8			
9	**Intermediate Calculations**		
10	Standard Error of the Mean	2.0000	= B7 / SQRT(B5)
11	Degrees of Freedom	24.0000	= B5 - 1
12	t test Statistic	0.5500	= (B6 - B2) / B10
13			
14	**Upper-Tail Hypothesis Test**		
15	Upper Critical Value	1.3178	= T.INV(1 - B4, B11)
16	p-value	0.2937	= 1 - T.DIST(ABS(B12), B11, TRUE)
17	Decision	**Do not reject null hypothesis**	= IF(B16 < B4, "Reject null hypothesis", "Do not reject null hypothesis")

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Hypothesis Tests for Proportions

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Hypothesis Tests for Proportions

- Involves categorical variables.
- Categorical variable must have only two possible outcomes.
  - Gender variable: Male or Female.
  - Coin variable: Head or Tail.
- Fraction or proportion of the population in the category of interest is denoted by π.

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Hypothesis Tests for Proportions

- Sample proportion in the category of interest is denoted by p.

$$p = \frac{X}{n} = \frac{\text{number in category of interest within sample}}{\text{sample size}}$$

- When both $n\pi$ and $n(1-\pi)$ are at least 5, the distribution of the sample proportion (p) can be approximated by a normal distribution with mean (μ) and standard deviation (σ) given by:

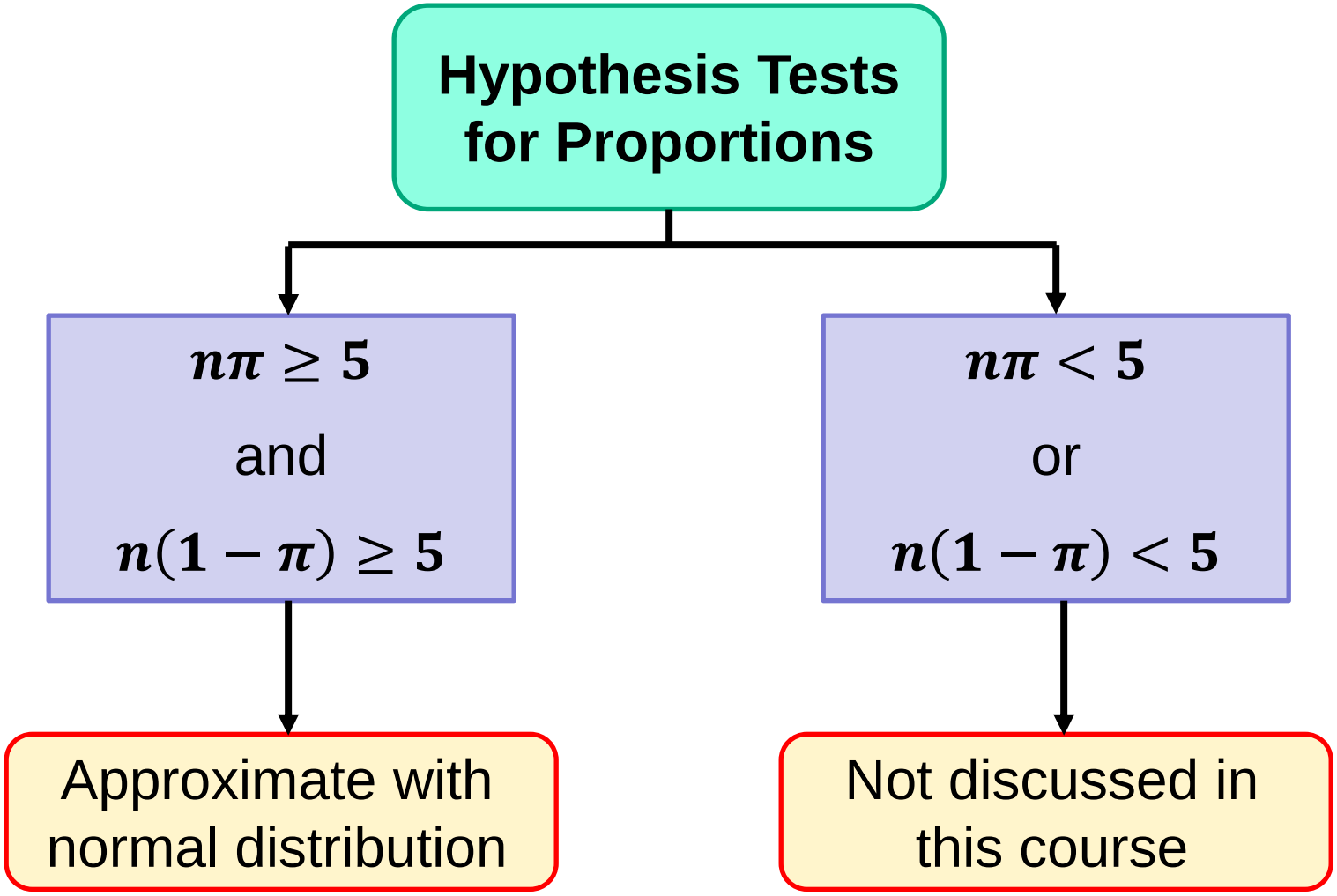
$$\mu = \pi$$

$$\sigma = \sqrt{\frac{\pi(1-\pi)}{n}}$$

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Hypothesis Tests for Proportions



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Hypothesis Tests for Proportions

- When the sampling distribution of p can be approximated with a normal distribution, the test statistic is given by the following expression.

$$z_{stat} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

- Since $x = np$, it implies that the expression of the test statistic given above can be re-expressed as:

$$z_{stat} = \frac{x - n\pi}{\sqrt{n\pi(1-\pi)}}$$

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Hypothesis Tests for Proportions

Example

A marketing company claims that it receives 8% responses from its mailing. To test this claim, a random sample of 500 subjects were surveyed and 25 responses were received. Verify the claim at the $\alpha = 0.05$ significance level.

Check

$$n\pi = 500 \times 8\% = 40$$

$$n(1-\pi) = 500 \times (1-8\%) = 460$$

Therefore, since $n\pi$ and $n(1-\pi)$ are both greater than 5, we can approximate with a normal distribution.

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Hypothesis Tests for Proportions

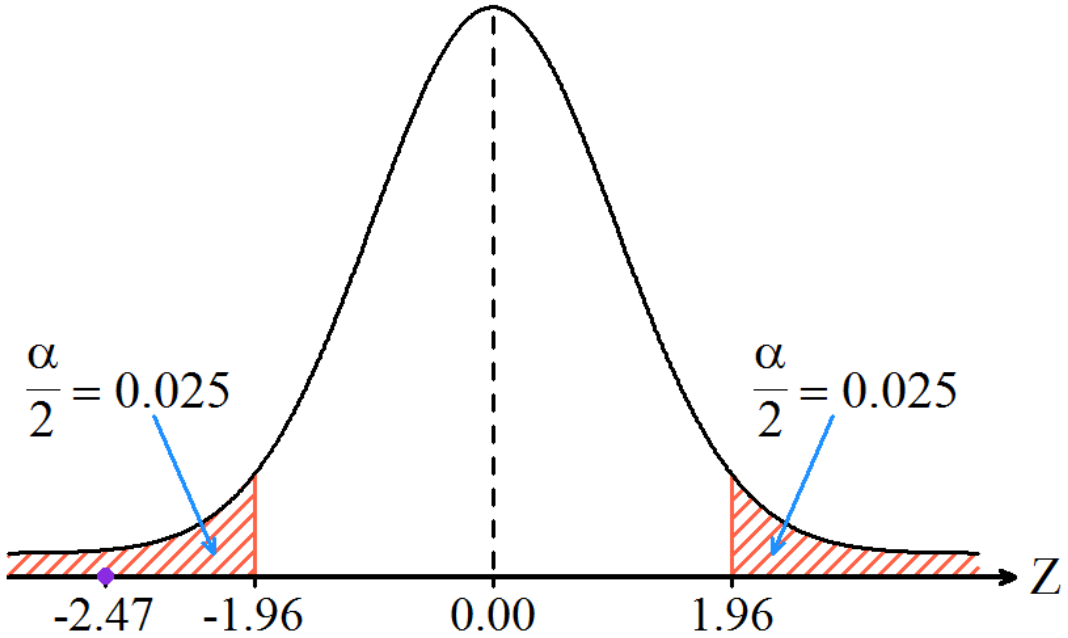
Solution (Critical Value Approach)

$\mathbf{H_0}: \boldsymbol{\pi} = \mathbf{8\%}$

$\mathbf{H_1}: \boldsymbol{\pi} \neq \mathbf{8\%}$

- $\alpha = 0.05.$
- $n = 500.$
- $p = \frac{25}{500} = 0.05.$
- Critical value:

$\pm z_{\alpha/2} = \pm z_{0.025} = \pm 1.96.$



$$z_{stat} = \frac{p - \pi}{\sqrt{\pi(1-\pi)/n}} = \frac{0.05 - 0.08}{\sqrt{0.08(1-0.08)/500}} = -2.47$$

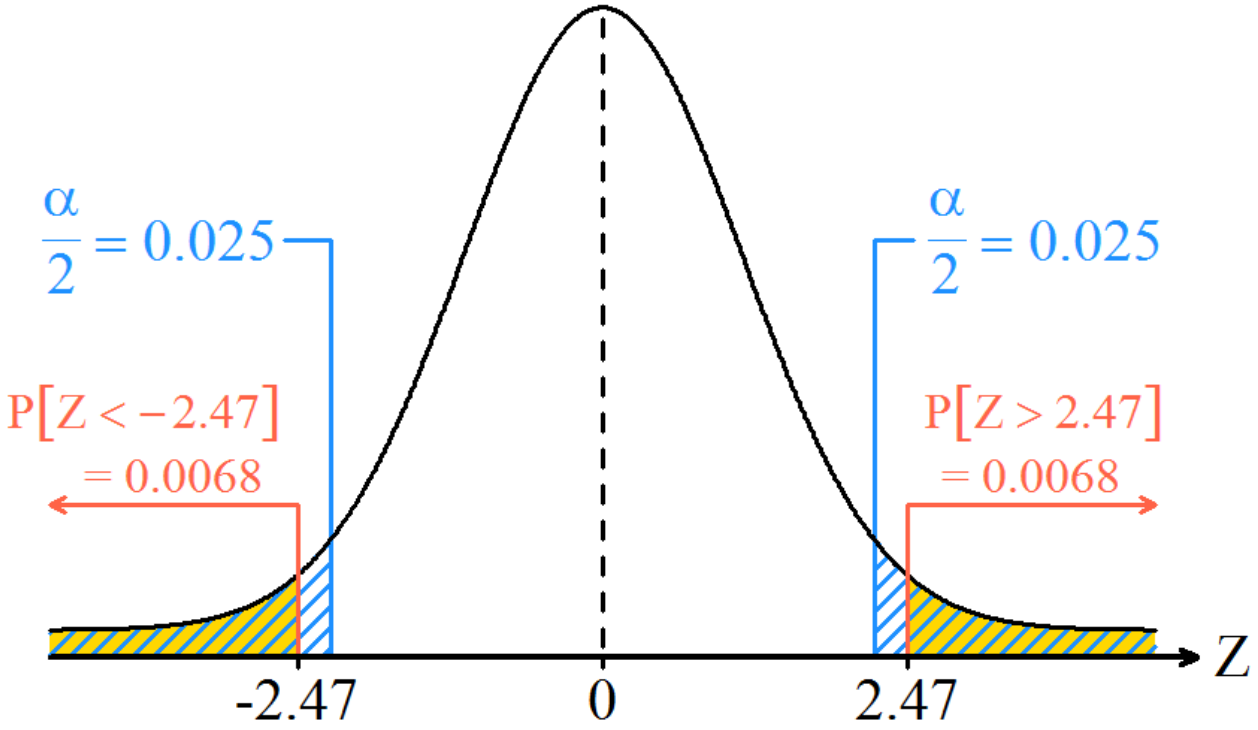
Reject $\mathbf{H_0}$: $-2.47 < -1.96$. Thus, the evidence from the data does not support the company's claim of 8% response rate.

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Hypothesis Tests for Proportions

Solution (p-value Approach)



$$z_{stat} = \frac{0.05 - 0.08}{\sqrt{0.08(1-0.08)/500}} = -2.47$$

$$p\text{−value} = \mathrm{P}(Z \le -2.47) + \mathrm{P}(Z \ge 2.47)$$
$$\approx 0.0068 + 0.0068 = 0.0136$$

Reject H_0: $0.0136 < \alpha = 0.05$. Therefore, the data contain sufficient evidence to reject the company's claim of 8% response rate.

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Questions to Address in the Planning Stage of Hypothesis Testing

- What is the goal of the survey, study or experiment?
- How can you translate the goal into a null and an alternative hypothesis?
- Is the hypothesis test one-tail or two-tail?
- Can a random sample be selected?
- What type of data set will be collected – numerical or categorical?
- What level of significance should be used?
- Is the intended sample size large enough to achieve the desired power?
- What statistical test procedure should be used and why?
- What conclusions and interpretations can be reached from the results of the planned hypothesis test?

Failure to consider these questions can lead to bias or incomplete results!.

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Statistical Significance vs Practical Significance

- Statistically significant results (that is, rejecting the null hypothesis) are not always of practical significance.
  - This is more likely to happen when the sample size gets very large.
- Practically important results might be found to be statistically insignificant (that is, failing to reject the null hypothesis).
  - This is more likely to happen when the sample size is relatively small.

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Reporting Findings and Ethical Issues

- Should document and report both good and bad results.
- Should not just report statistically significant results.
- Reports should distinguish between poor research methodology and unethical behaviour.
- Ethical issues can arise so they must be properly handled and reported. These include,
  - The use of human subjects.
  - The data collection method.
  - The type of test being used.
  - The level of significance being used.
  - The cleaning and discarding of the data.
  - The failure to report pertinent findings.

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Power of the Test

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Power of the Hypothesis Test

- The power of an hypothesis test $(1 - \beta)$ is the probability of rejecting a false null hypothesis.
  - $\boldsymbol{\beta}$: the probability of Type II Error. That is, the probability of failing to reject a false H_0.

- Power of the test depends on the following.
  - The magnitude of the difference between the hypothesised and the true mean value.
  - The size of the significance level (α).
  - Whether the test is one or two-tailed.

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Power of the Hypothesis Test

One-tail test is more powerful compared to two-tailed test.

The higher the difference between the true and hypothesised means, the higher the power of the test.

Higher significance level (α) causes the power of the test to increase.

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Power of the Hypothesis Test

Example

A cereal filling machine is designed to discharge an average of 368g per box with a standard deviation (σ) of 15g. An inspector from the consumer protection unit is interested in validating a complaint that the machine dispenses less than the advertised quantity. If a random sample of 25 boxes was taken and a significance level $\alpha = 0.05$ is adopted, compute the power of the test if the actual amount of cereal discharged is 360g and σ remains 15g.

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Power of the Hypothesis Test

Solution

1. Calculate the critical value that corresponds to the specified significance level, under the "assumed" distribution.

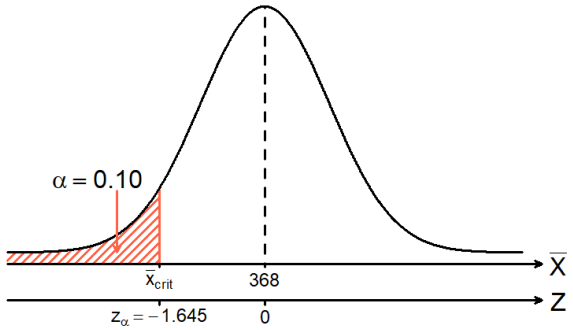
$$z_\alpha = z_{0.05} = \frac{\bar{x}_{crit} - \mu}{\sigma/\sqrt{n}}$$

$$-1.645 = \frac{\bar{x}_{crit} - 368}{15/\sqrt{25}}$$

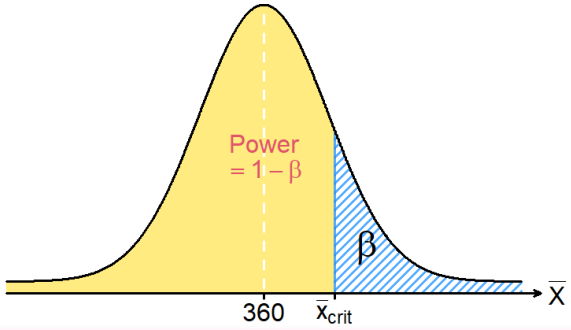
$$\bar{x}_{crit} = 368 - \frac{1.645 \times 15}{\sqrt{25}}$$

$$\approx 363.0650$$

Assumed Distribution



True Distribution

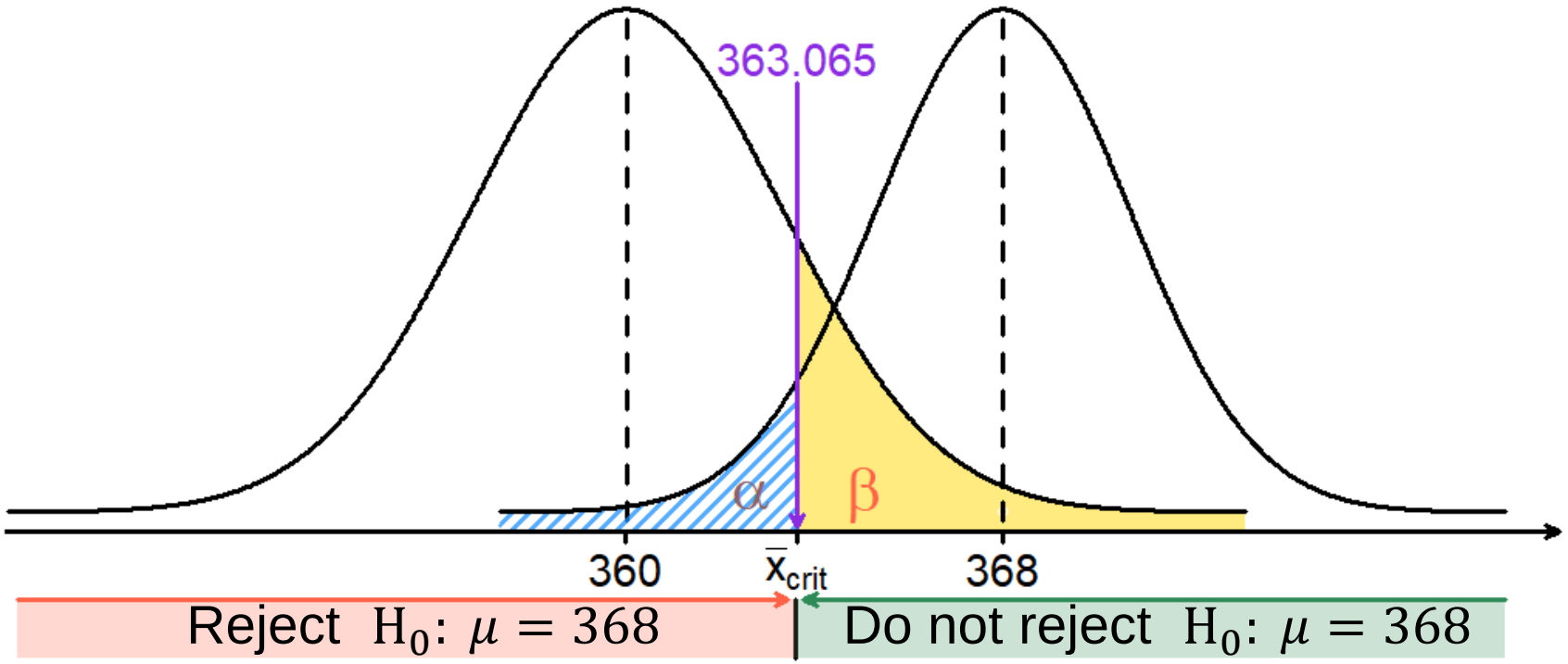


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Power of the Hypothesis Test

Solution

2. Determine $\beta = \mathrm{P}(\text{Type II Error})$. That is, the probability of observing $\bar{x}_{crit}$ or more extreme values under the **true** distribution.



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Power of the Hypothesis Test

Solution

2. Determine $\beta = \mathrm{P}(\text{Type II Error})$. That is, the probability of observing $\bar{x}_{crit}$ or more extreme values under the **true** distribution.

$$\beta = \mathrm{P}(\bar{X} > \bar{x}_{crit} \mid \mu = 360)$$

$$z_{stat} = \frac{\bar{x}_{crit} - 360}{15/\sqrt{25}}$$

$$= \frac{363.0650 - 360}{15/\sqrt{25}} \approx 1.0217$$

$$\therefore \beta = \mathrm{P}(Z > 1.0217)$$

$$= 1 - \mathrm{P}(Z < 1.0217)$$

$$\approx 1 - 0.8461 = 0.1539$$

$$\Rightarrow \text{Power} = 1 - \beta$$

$$= 1 - 0.1539 = \mathbf{0.8461}.$$

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Chapter Summary

In this chapter, we discussed:

- The principles of hypothesis testing.
- How to use hypothesis testing to test a mean or proportion.
- One-tail vs Two-tail tests and Critical value vs p-value approach to hypothesis testing.
- The pitfalls and ethical issues involved in hypothesis testing and how they can be avoided.
- How to evaluate power of the test.

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